

PCS-302 : Data Structures Lab

Stack Using an Array | Create, Push, Pop, IsEmpty, Size, Print

Problem Statement

Design an algorithm and a program to implement a stack using an array. The program should implement the following stack operations: Create(), Push(k), Pop(), IsEmpty(), Size() and Print().

Example

```
create()    -->  Stack created, empty

push(10)    -->  Stack: 10
push(20)    -->  Stack: 10 20
push(30)    -->  Stack: 10 20 30

size()      -->  3

pop()       -->  removes 30
print()     -->  20 10    (top to bottom)
```

Algorithm

```
Create():
    top = -1                                // empty stack

Push(k):
    if top == MAX - 1: print "Overflow", return
    top = top + 1
    stack[top] = k

Pop():
    if top == -1: print "Underflow", return
    remove stack[top]
    top = top - 1

IsEmpty():
    return (top == -1)

Size():
    return (top + 1)

Print():
    for i = top down to 0:
        print stack[i]
```

C Program

```
#include <stdio.h>
#define MAX 5

int stack[MAX];
int top = -1;
```

```

void create() {
    top = -1;                                // empty stack
    printf("Stack created.\n");
}

int isEmpty() {
    if (top == -1)
        return 1;
    else
        return 0;
}

void push(int k) {
    if (top == MAX - 1) {
        printf("Stack Overflow. Cannot push %d.\n", k);
        return;
    }
    top = top + 1;
    stack[top] = k;
    printf("%d pushed to stack.\n", k);
}

void pop() {
    if (isEmpty() == 1) {
        printf("Stack Underflow. Cannot pop.\n");
        return;
    }
    printf("%d popped from stack.\n", stack[top]);
    top = top - 1;
}

int size() {
    return top + 1;
}

void print() {
    int i;
    if (isEmpty() == 1) {
        printf("Stack is empty.\n");
        return;
    }
    printf("Stack (top to bottom): ");
    for (i = top; i >= 0; i--) {
        printf("%d ", stack[i]);
    }
    printf("\n");
}

int main() {
    create();

    push(10);
    push(20);
    push(30);
    print();
}

```

```
printf("Size = %d\n", size());

pop();
print();

if (isEmpty() == 1)
    printf("Stack is empty.\n");
else
    printf("Stack is not empty.\n");

return 0;
}
```